

```
In [1]: #Disctionary
```

```
In [4]: dictionary ={}  
1
```

```
Out[4]: {}
```

```
In [5]: type(dictionary)
```

```
Out[5]: dict
```

```
In [ ]: s = {10,20,30}
```

```
In [ ]: type(s)
```

```
In [6]: d={1:'one', 2:'two', 3: 'three'}  
type(d)
```

```
Out[6]: dict_items([(1, 'one'), (2, 'two'), (3, 'three')])
```

```
In [7]: d.keys()  
d.values()  
d.items()
```

```
Out[7]: dict_items([(1, 'one'), (2, 'two'), (3, 'three')])
```

```
In [8]: d[4]='Four'
```

```
In [9]: d
```

```
Out[9]: {1: 'one', 2: 'two', 3: 'three', 4: 'Four'}
```

```
In [11]: d[5]='five'  
d
```

```
Out[11]: {1: 'one', 2: 'two', 3: 'three', 4: 'Four', 5: 'five'}
```

```
In [16]: keys = {'a','b','c','d'}  
mydict3 = dict.fromkeys(keys)  
mydict3
```

```
Out[16]: {'a': None, 'c': None, 'b': None, 'd': None}
```

```
In [17]: keys = {'a','b','c','d'}  
value = 10  
mydict3 = dict.fromkeys(keys,value)  
mydict3
```

```
Out[17]: {'a': 10, 'c': 10, 'b': 10, 'd': 10}
```

```
In [18]: d
```

```
Out[18]: {1: 'one', 2: 'two', 3: 'three', 4: 'Four', 5: 'five'}
```

```
In [20]: d[1]
```

```
Out[20]: 'one'
```

```
In [21]: d[5]
```

```
Out[21]: 'five'
```

```
In [22]: d.get(2)
```

```
Out[22]: 'two'
```

```
In [23]: d.pop(5)  
d
```

```
Out[23]: {1: 'one', 2: 'two', 3: 'three', 4: 'Four'}
```

```
In [27]: #RANGE
```

```
In [28]: range(5) # Range with one argument
```

```
Out[28]: range(0, 5)
```

```
In [29]: list(range(0,5))
```

```
Out[29]: [0, 1, 2, 3, 4]
```

```
In [32]: for i in range(0,5) :  
         print(i)
```

```
0  
1  
2  
3  
4
```

```
In [33]: range(10,20)
```

```
Out[33]: range(10, 20)
```

```
In [34]: list(range(20))
```

```
Out[34]: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19]
```

```
In [36]: list(range(0,20,5))
```

```
Out[36]: [0, 5, 10, 15]
```

```
In [37]: range(0,10,2,3) # Range take max 3 arguments
```

TypeError

Traceback (most recent call last)

Cell In[37], line 1

----> 1 range(0,10,2,3)

TypeError: range expected at most 3 arguments, got 4

```
In [ ]: range(0,10,2,3)
```

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In [ ]:
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In [ ]:
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In [ ]:
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